Zhi CHEN (陈智)

Ph.D. Student · Intelligent Software Engineering Researcher

zhi.chen.2023@phdcs.smu.edu.sg | LinkedIn | Google Scholar | Homepage

Summary

With a consistent B.Eng., M.Sc., and Ph.D. focus in Software Engineering, I have built strong foundations and gained two years of full-time industry experience as a software engineer. Currently a PhD student, I focus on research of practical Code AI systems, including Large Code Models and Software Engineering Code Agents, bridging academic insights with real-world applications.

Education

Singapore Management University — *Singapore*

Ph.D. in Computer Science — *Aug. 2023 – Present*

- Centre for Research on Intelligent Software Engineering (RISE Centre)
- Advisor: Prof. Lingxiao Jiang (Homepage)
- Research Area: AI-Driven Software Development, Software Engineering Agents, Large Code Models

Singapore Management University — *Singapore*

M.Sc. in Computing (Track: Software & Cyber-Physical Systems) — *Aug. 2021 – Aug. 2022*

- School of Computing and Information Systems
- GPA: 3.80/4

Hefei University of Technology — *China*

B.Eng. in Software Engineering — *Sep. 2016 – Jul. 2020*

- School of Software
- Grade: 85.07/100

National Yunlin University of Science & Technology — *Taiwan*

Exchange Student — *Feb. 2018 – Jun. 2018*

- Department of Computer Science and Information Engineering
- GPA: 3.68/4

Work Experience

TikTok — *Singapore*

Research & Development Intern – AI Innovation Center — *Dec 2025 – Present*

- **Project:** Multi-Agent Systems for Code Optimization
- **Motivation:** Large-scale production systems serve hundreds of millions of users, where inefficient code increases CPU usage and operational cost. Improving code efficiency reduces CPU time and lowers service cost.
- **Outcome:** Built an end-to-end multi-agent system for code optimization, covering hotspot localization, optimization generation, and performance evaluation.
- **Contribution:** Worked across the full end-to-end optimization pipeline, contributing to hotspot identification, optimization patch generation, CPU reduction evaluation, service-team coordination, patch deployment, and validation of production performance gains.

Huawei — *Singapore*

Research Intern – Noah's Ark Lab — *Apr. 2025 – Aug. 2025*

- **Project:** Deep-Research Agent – Knowledge Integration and Literature Review Assistant
- **Motivation:** LLMs are limited by their internal knowledge when answering questions; integrating external search engines allows access to up-to-date information, improving answer quality and reducing hallucinations.
- **Outcome:** A modular LangGraph-based agent capable of integrating web and ArXiv search for knowledge synthesis; a proof-of-concept literature review generator; co-authored a survey on Reinforcement Learning for Deep Research Agents.
- **Contribution:** Refactored original Python demo to LangGraph architecture for better maintainability; integrated web and ArXiv search engines; implemented multi-step retrieval and summarization pipelines; contributed to survey writing.

Sea Limited — *Singapore*

Software Engineer – Sea Labs — *Aug. 2022 – Jul. 2023*

- **Project:** "Loop" – Survey Platform for Product Incubation
- **Motivation:** Support early-stage product development by collecting structured feedback from employees, enabling product managers to make informed decisions for product improvements.
- **Outcome:** Successfully launched a production-ready survey platform from 0 to 1, enabling PMs to collect and analyze structured feedback at scale; established CI/CD pipeline for continuous deployment.
- **Contribution:** Developed server-side modules (Authentication, Survey Config, Notification, Cron), implemented efficient data aggregation with goroutines and Redis caching, and visualized survey insights to guide PM decisions.

Shanghai Baosight Software Co., Ltd. *Shanghai, China*

Software Engineer – Cloud Application Dept. *Jul. 2020 – Jul. 2021*

- **Project:** Blockchain-based Supply Chain Finance System
- **Motivation:** Enhance transparency and traceability in warehouse receipt financing using blockchain.
- **Outcome:** Delivered a secure system enabling tokenized warehouse receipts and traceable logistics operations; smart contracts successfully deployed in Ethereum-compatible blockchain.
- **Contribution:** Implemented asynchronous blockchain callbacks and certification generation module, ensuring secure and auditable financial transactions.

Lenovo Group, LCFC (Hefei) Electronics Technology Co., Ltd. *Hefei, China*

Software Engineer Intern – R&D Department *Oct. 2019 – Feb. 2020*

- **Project:** Model-Based Inspection System
- **Motivation:** Automate appearance quality control across ThinkPad product lines.
- **Outcome:** Delivered web APIs and frontend dashboards enabling systematic management of visual inspection datasets.
- **Contribution:** Migrated core inspection system from .NET to Java-based architecture, improving maintainability and scalability.

Publications

Are Performance-Optimization Benchmarks Reliably Measuring Coding Agents? [PDF] *ICSE 2027*

First-Author

Zhi Chen, Zhensu Sun, Yuling Shi, David Lo, Lingxiao Jiang. 49th IEEE/ACM International Conference on Software Engineering (ICSE 2027).

Beyond Final Code: A Process-Oriented Error Analysis of Software Development Agents in Real-World GitHub Scenarios [PDF] *ICSE 2026*

First-Author

Zhi Chen, Wei Ma, Lingxiao Jiang. 48th International Conference on Software Engineering (ICSE 2026).

Evaluating Software Development Agents: Patch Patterns, Code Quality, and Issue Complexity in Real-World GitHub Scenarios [PDF] *SANER 2025*

First-Author

Zhi Chen, Lingxiao Jiang. 32nd IEEE International Conference on Software Analysis, Evolution, and Reengineering (SANER 2025).

Promise and Peril of Collaborative Code Generation Models: Balancing Effectiveness and Memorization [PDF] *ASE 2024*

First-Author

Zhi Chen, Lingxiao Jiang. 39th IEEE/ACM International Conference on Automated Software Engineering (ASE 2024).

Rethinking the Value of Agent-Generated Tests for LLM-Based Software Engineering Agents [PDF] *Under Review*

First-Author

Zhi Chen, Zhensu Sun, Yuling Shi, Chao Peng, Xiaodong Gu, David Lo, Lingxiao Jiang.

ConfuScope: A Multi-Agent Framework via Hybrid Search and Content-Aware Analysis for Robust Package Confusion Detection [PDF] *TSE 2026*

Third-Author

Yu Li, Wei Ma, Zhi Chen, Ye Liu, Lingxiao Jiang, Junyi Tao, Hao Liu, Yongqiang Lyu, Qiang Hu. IEEE Transactions on Software Engineering (TSE).

AgentGuard: An Active Threat Discovery System for Package Confusion Using Multi-Agent Collaboration [PDF] *ML4CS 2025*

Third-Author

Wei Ma, Yu Li, Zhi Chen, Ye Liu, Lingxiao Jiang, Qiang Hu, Junyi Tao. 7th International Conference on Machine Learning for Cyber Security (ML4CS 2025).

Same Signal, Different Semantics: A Cross-Framework Behavioral Analysis of Software Engineering Agents [PDF] *arXiv preprint*

Second-Author

Wei Ma, Zhi Chen, Jingxu Gu, Tianling Li, Shangqing Liu, Lingxiao Jiang.

Executing as You Generate: Hiding Execution Latency in LLM Code Interpreters [PDF] *arXiv preprint*

Third-Author

Zhensu Sun, Zhihao Lin, Zhi Chen, Chengran Yang, Mingyi Zhou, Li Li, David Lo.

Reinforcement Learning Foundations for Deep Research Systems: A Survey [PDF] *arXiv preprint*

Second-Author

Wenjun Li, Zhi Chen, Jingru Lin, Hannan Cao, Wei Han, Sheng Liang, Zhi Zhang, Kuicai Dong, Dexun Li, Chen Zhang, Yong Liu.

Honors & Awards

2026	**Nominated for Global Young Scientists Summit (GYSS) 2027**, Singapore Management University	*Singapore*
2026	**SMU Presidential Doctoral Fellowship**, Singapore Management University	*Singapore*
2025	**SMU Presidential Doctoral Fellowship**, Singapore Management University	*Singapore*
2023	**SMU Research Scholarship**, Singapore Management University	*Singapore*
2021	**Merit Award**, Singapore Management University	*Singapore*
2018	**Golden Prize in Fifth Internet Innovation Competition**, Hefei University of Technology	*Anhui, China*
2017	**Second-Class Merit-Based Scholarship**, Hefei University of Technology	*Anhui, China*

Academic Service

Reviewer: ACM Transactions on Software Engineering and Methodology (TOSEM)